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Digital Prayer Beads: Adaptive Kinesthetic Mindfulness Device for Emotion Regulation and Stress Relief

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ABSTRACT
In light of the growing focus on youth mental health, the exploration of effective emotion regulation strategies is of paramount significance. This study designs and evaluates a novel kinesthetic mindfulness device, Digital Prayer Beads, which supports standard mindfulness practice through intuitive interactive hand movements to enhance users' non-judgmental present-moment awareness. Based on a targeted user requirement survey of college students and working professionals, the device was prototyped and systematically validated. Thirty participants were randomly divided into natural recovery, auditory feedback, and audiovisual feedback groups. Combining subjective anxiety assessments and objective physiological indicators including skin conductance response and heart rate variability, rigorous statistical analyses were conducted. The results confirm the device effectively alleviates user anxiety, with audiovisual feedback achieving optimal effects. It provides empirical evidence for kinesthetic interaction-based emotion regulation and offers a novel approach to integrate mindfulness practices into youth mental health promotion.

KEYWORDS
Kinesthetic mindfulness; emotion regulation; stress relief; digital prayer beads; adaptive feedback

1. Introduction

In recent years, global mental health issues have gained increasing attention, especially among youth facing high work demands and academic competition, with anxiety and depression rates steadily rising (Chodavadia et al., 2023; B. Lu et al., 2024; Y. Sun et al., 2023). Data projected by the World Health Organization (WHO) for 2025 indicate that over 1 billion people worldwide are affected by mental disorders, with anxiety and depression being the most prevalent (World Health Organization, 2022). These conditions not only result in significant human losses but also impose a substantial economic burden, with depression and anxiety alone causing an estimated annual global loss of approximately 1 trillion USD (World Health Organization, 2023). In China, the lifetime prevalence rate of mental disorders among adults is 15.87% (L. Sun et al., 2021), while adolescent mental health issues are becoming more common and manifest at younger ages. For example, the "2022 China Youth Mental Health Report" shows that 14.8% of adolescents are at risk of depression (Guo et al., 2023). Of particular concern is the fact that the prevalence of depression and anxiety symptoms among university students stands at 33.6 and 39.0%, respectively, with academic pressure, family relationships, and interpersonal issues being the main contributing factors (Deng et al., 2022; W. Li et al., 2022). At the same time, various professional groups are also encountering significant mental health challenges due to factors such as high work intensity, digital fatigue, and isolation from their environments (Arıcan, 2025; Arican & Ünal, 2023).

Despite cognitive-behavioral therapy and pharmacological treatments being the mainstream approaches, many individuals face barriers to accessing timely and effective help due to high costs, lengthy treatment durations, medication side effects, and societal stigma (Basile et al., 2024; Braun et al., 2023; Issac et al., 2025; L. Williams et al., 2024). Simultaneously, existing mass-market self-help tools, such as meditation apps, primarily rely on visual and auditory channels. In high-pressure situations where cognitive resources are already saturated, these tools may increase sensory overload, and

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their "one-size-fits-all" approach makes it difficult to cater to the diverse emotional regulation needs of different users (Hwang et al., 2022; Shi et al., 2023). To address these challenges, there is an urgent need to develop non-pharmacological emotional regulation tools that are low in cognitive demand, easily accessible, and seamlessly integrated into daily life.

The tactile sensory channel, due to its intuitiveness, low invasiveness, and strong emotional connection (Rognon et al., 2022; Vasseur et al., 2024), offers unique benefits for promoting mindfulness practices, which involve anchoring the present moment and enhancing awareness through bodily sensory experiences (Djernis et al., 2023). "Dynamic mindfulness anchoring," based on repetitive hand movements such as manipulating prayer beads, can effectively maintain attention on bodily sensations and interrupt rumination, aligning with the core principles of mindfulness practice (Bolzenkötter et al., 2025). However, most current tactile intervention products are limited to providing passive feedback, lacking user control and personalized involvement, which hinders the effectiveness of emotional regulation (Ban et al., 2022; Haynes et al., 2022; MacDonald et al., 2024).

Integrating the empirical research and theoretical findings on tactile-mediated emotional regulation, kinesthetic mindfulness grounding, and traditional dynamic meditation, we combine the interactive principles of traditional mindfulness tools with modern human-computer interaction technologies. This results in the development of an interactive emotional regulation system, which focuses on hand-based tactile movements, as illustrated in Figure 1. The design builds upon the body-mind integration mechanism of traditional meditation beads (Crosswell et al., 2024), where users maintain dynamic focus through active physical engagement, such as turning or spinning the beads. This, in combination with multimodal sensors and a feedback system, creates a closed-loop "operation-feedback" interaction. The system provides a low-cognitive-load emotional self-help tool for young people facing high levels of stress in their daily lives.

The primary contribution of this study lies in the design of an innovative digital kinesthetic mindfulness interactive device, which integrates hand gestures and multimodal feedback to assist with emotional regulation and stress relief. By combining mindfulness practices with contemporary human-computer interaction technologies, this study examines the effect of continuous manual interactions on emotional regulation. Through prototype design and validation, the study assesses the practical feasibility of implementing the device. These contributions offer a new perspective on the design of digital emotional regulation tools based on bodily movements and provide a solid foundation for further research in related fields.

The main research questions (RQs) addressed in this study are:

RQ1: How can a kinesthetic mindfulness device be designed for emotional regulation?

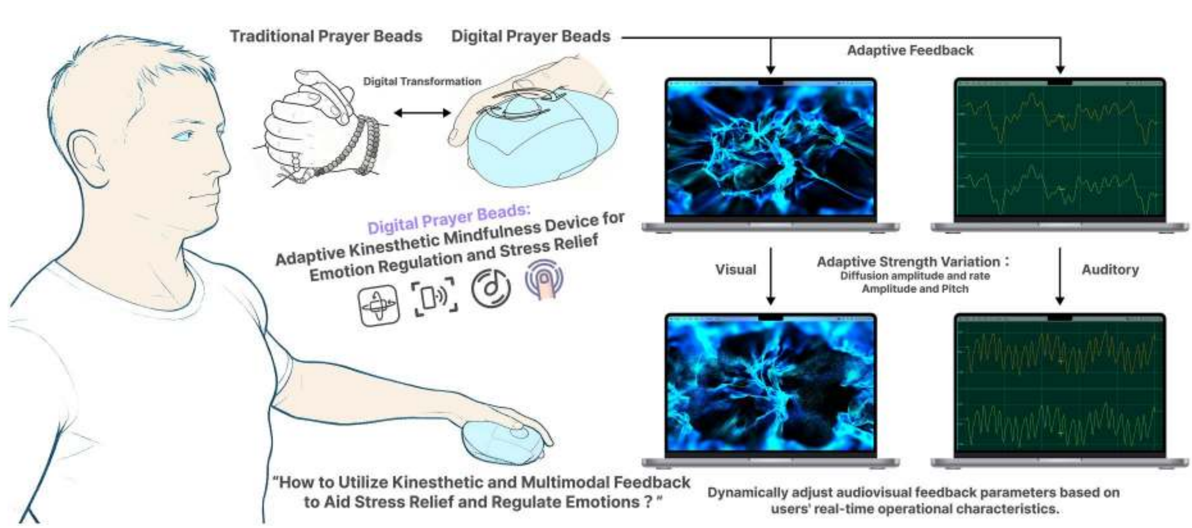

Figure 1. A kinesthetic mindfulness device for emotional regulation: Integrating the concept of prayer beads with behavioral actions to deliver adaptive kinesthetic mindfulness for emotional regulation and stress relief. *Note:* Designed by the authors.

RQ2: How do the multimodal feedback components in the digital prayer beads work together to regulate the user's emotional state?

RQ3: What are the subjective experiences, usability, and stress-relief effects of users when interacting with the digital prayer beads device?

RQ4: How does the digital prayer beads device influence users' attention focus and present-moment awareness?

This section aims to outline the theoretical foundation of the study, identify the gaps in existing research, and lay the groundwork for the subsequent design. It will systematically review relevant research outcomes in related fields, forming the "Related Work" section.

2. Related work

2.1. The significance of prayer beads

In Buddhism, prayer beads are a significant tool for spiritual practice, deeply embedded in historical and practical traditions. The use of prayer beads traces back to ancient India's religious practices. Since the 5th century BCE, Buddhism has adopted and formalized their use, assigning them the purpose of eliminating mental afflictions. The earliest documented reference to prayer beads appears in the *Mulian Sutra*, where Buddha instructed King Boli to create 108 beads from the soapberry tree (Sapindus) to count and recite the Buddha's name, thus purging afflictions (Dorji, 2025). The spread of prayer beads in China is tied closely to the rise of the Pure Land sect, with Master Daochuo contributing by inventing the "counting beans" method and using soapberry wood, making prayer beads a widely practiced tool in Buddhist rituals.

The role of prayer beads in spiritual practice extends beyond mere counting to integrate the "body, speech, and mind" in meditation. Through focused recitation, practitioners seek spiritual transcendence. The use of beads is believed to eliminate both mental and karmic obstacles, and reciting a mantra 108,000 times is considered a complete spiritual practice (Dorji, 2025). The number of beads often has symbolic meaning, such as 108 beads symbolizing the elimination of 108 types of afflictions. By engaging with the beads, practitioners maintain focus and experience bodily resonance, achieving spiritual transcendence. Thus, the profound symbolic and spiritual value of prayer beads is an essential aspect of Buddhist practice (Markum et al., 2024).

However, the existing literature lacks investigations into the digital transformation and interactive design of traditional Buddhist prayer beads. Few studies have explored how to integrate the symbolic value and kinesthetic practice of prayer beads into modern digital mental health interventions for young people in daily life.

2.2. Kinesthetic perception in mindfulness practice and traditional tools

Mindfulness refers to the practice of being aware of the present moment in a conscious and non-judgmental manner, with its roots in Eastern Buddhist meditation traditions (Chems-Maarif et al., 2025). In modern psychology, it has evolved into a series of structured interventions. Among these, Mindfulness-Based Stress Reduction (MBSR) and Mindfulness-Based Interventions (MBI) have been widely studied and proven to be effective psychological practices for reducing stress, anxiety, and improving emotion regulation (Abdul Manan et al., 2024; Hoge et al., 2023; Sercekman & Meltem, 2024). The core components of mindfulness are typically described in terms such as observation, description, purposeful action, non-reactivity, and non-judgment (Heeren et al., 2021; Takahashi et al., 2022). Notably, there are significant differences in the methods of mindfulness practice across cultures: Eastern traditions particularly emphasize the integration of body movements and sensory experiences, while Western mindfulness interventions tend to focus more on cognitive awareness (Du & Ning, 2024; Galante et al., 2023).

In Eastern cultures, various traditional spiritual tools combine body movements with mindfulness anchoring. For instance, in Chinese Buddhism and Japanese Zen, practitioners use "prayer beads"

(Buddha beads), performing the ritual act of moving the beads with their fingers while reciting mantras. This practice helps maintain focus and calm the mind (Ouyang & Xie, 2025; Tonuk, 2020). Similarly, Tibetan Buddhism uses "prayer wheels" that accumulate merit through rotation, with the rhythmic motion aiding concentration (Brox, 2022). This approach of assisted spiritual practice via bodily rituals is also found in other religious traditions, such as the Islamic "Tasbih" and the Catholic "Rosary." Both use repetitive hand movements synchronized with prayer to enhance inner focus and tranquility (Kabir et al., 2024; Stöckigt et al., 2021; Teut et al., 2025). The common feature of these tools is that they anchor the practitioner's attention to the present moment through controlled, rhythmic bodily movements, promoting "mind-body unity" and emotional balance.

A well-known example of Eastern, body-participatory mindfulness is the "Baoding ball" from China, which has a long history and is recognized as a popular folk fitness tool (Peterson, 2024). Practitioners rotate a pair of metal balls in their palms, which contain sound plates that emit a crisp, harmonious sound when rotated. According to traditional Chinese medicine theory, moving the balls with the fingers stimulates acupoints in the palms, which helps open the meridians, regulate qi and blood, and calm the mind (Mould, 1996; A. Williams, 1997). From a mindfulness perspective, skillfully manipulating the Baoding balls requires great concentration and coordination. This process is a form of "dynamic meditation" that naturally directs attention away from distractions to physical sensations, promoting relaxation. Similarly, the repetitive hand movements involved in using prayer beads have been shown to slow heart rate and induce calm, with the core mechanism being the focus on tactile sensations and movement, which helps break the cycle of ruminative thinking and stabilize awareness in the present (Wernik, 2009).

In contrast, mainstream Western mindfulness interventions (such as MBSR and Mindfulness-Based Cognitive Therapy) also stem from Eastern meditation practices, but in the process of standardization and de-contextualization, they tend to emphasize practices like seated meditation, body scans, and breath awareness, with less use of tactile assistive tools found in traditional methods (Bondolfi et al., 2010; Keng et al., 2011).

Studies indicate that the effectiveness of tools like prayer beads and Baoding balls largely stems from the creation of an action-perception feedback loop. Emami et al.'s review also highlights that tactile experience plays a key role in traditional relaxation practices (Emami et al., 2025). For example, mastering the technique of using prayer beads or rotating the Baoding balls requires practitioners to focus on subtle motor control, which can naturally induce a flow state similar to deep concentration in meditation, rather than passively following external instructions. This suggests that rhythmic, self-controlled physical movements can more naturally trigger physiological and psychological relaxation responses.

However, the existing literature lacks systematic research on integrating traditional kinesthetic mindfulness tools with adaptive digital interactive technologies, especially designs that combine hand movement, real-time multimodal feedback, and personalized regulation for young people. In addition, few studies have empirically examined the stress relief and emotion regulation effects of such digitally transformed kinesthetic mindfulness devices using both psychological scales and physiological indicators.

2.3. Digital mindfulness tools and emotionally mindful touch

With the advancement of wearable technology and mobile computing, digital mindfulness tools are evolving from relying solely on audiovisual guidance to incorporating multimodal interactions (J. Li et al., 2024). However, many current mainstream applications, such as Headspace and Calm, still primarily rely on audiovisual cues. The potential of the tactile channel in emotional regulation, particularly emotionally mindful touch, remains underexplored and underutilized (Callahan et al., 2024; He et al., 2024; Macrynikola et al., 2024; Paterson, 2025).

From a neuroscientific perspective, human touch is functionally differentiated: besides the discriminatory touch responsible for identifying the physical properties of objects (such as shape and texture), there is an emotional touch system designed to convey social emotions and induce pleasant feelings (Pereira et al., 2025; Schirmer et al., 2023). This emotional touch system is mainly mediated by a neural pathway called C-tactile afferent fibers, which are most sensitive to slow, gentle touches at near-body temperatures, and can directly transmit such tactile signals to brain regions associated with emotional processing, such as the insula (R. Liu & Kondo, 2025; Papi et al., 2025). This mechanism provides a theoretical foundation for mindfulness interaction models that can trigger positive emotional responses and foster inner calm (Birznieks, 2025; Schlintl & Schienle, 2024).

In the realm of human-computer interaction, some studies have started to explore the emotional dimension of mindful touch (Cheung et al., 2025). For instance, Haynes et al. noted that traditional applications overly rely on audiovisual cues, often neglecting tactile guidance. They developed a novel haptic interface that simulates slow breathing using pneumatic systems (Haynes et al., 2022). Similarly, a breathing pillow that mimics the tactile rhythm of slow breathing has been shown to effectively reduce anxiety levels, aligning with the emotional touch principle of providing comfort through rhythmic tactile stimulation (Theodore et al., 2024). Moreover, active exploration of guided imagery through actions (such as grasping virtual objects) has been shown to influence the mindfulness experience via multimodal interactions (Mancini et al., 2024). Products like Sensate and Doppel are also available, using gentle vibrations on the chest, back, or wrist to regulate users' physiological rhythms and reduce anxiety (Motti & Faiz, 2025; T. Azevedo et al., 2017).

Despite these advancements, existing designs still have significant limitations: most products emphasize passive feedback reception, with limited opportunities for active bodily engagement from users. Moreover, the design of tactile feedback does not fully optimize the emotional touch activation pattern. An effective digital meditation tool design should better integrate the neural mechanisms of emotional touch with the concept of "mindfulness grounding." This includes exploring how users can actively engage with the tool through controllable bodily movements (rather than passively receiving feedback) while receiving dynamic feedback that aligns with the characteristics of emotionally mindful touch. This approach would more effectively help users anchor their attention to the present bodily experience and promote deep emotional regulation. The design should create a closed-loop interaction where actions and feedback are tightly connected, making touch not just a sensory output but a core channel that carries emotional meaning and supports mindfulness practice.

However, the existing literature lacks research on active kinesthesia-driven affective tactile mindfulness devices. Few designs combine real-time physiological signals with adaptive multimodal tactile feedback to support personalized emotion regulation, which limits the depth and sustainability of mindfulness intervention effects.

2.4. Psychological mechanisms of immersive interaction and emotion regulation

In the field of human-computer interaction research for emotion regulation, immersion and engagement are considered critical mechanisms for enhancing the effectiveness of interventions (Raman et al., 2025). These concepts are rooted in flow theory, as well as other psychological frameworks such as Self-Determination Theory (SDT) and the Proteus Effect (Lin et al., 2025; Roh et al., 2024; Tyack & Mekler, 2024).

According to flow theory, individuals are more likely to enter a state of deep concentration when the challenge of a task aligns with their skills, goals are clear, and feedback is immediate (Wojtasiński et al., 2025). In interaction design, fostering flow states relies heavily on providing users with active control and closed-loop feedback. For instance, certain studies have demonstrated that visualizing breathing exercises can enhance learning outcomes, while combining breathing biofeedback with full-body avatars to offer VR-based breathing exercises rooted in mindfulness principles shows that active interaction, coupled with multi-sensory feedback, can optimize emotional regulation (Ng et al., 2024; Yang et al., 2025).

Self-Determination Theory, developed by Deci and Ryan, emphasizes the importance of fulfilling three core psychological needs—autonomy, competence, and relatedness—in fostering intrinsic motivation (Deci & Ryan, 2012). In interaction design, providing users with a sense of control (autonomy), offering clear and timely feedback (competence), and creating emotional resonance (relatedness) are key to enhancing the depth and sustainability of emotion regulation efforts (Alberts et al., 2026; Bennett & Mekler, 2024; Cotter et al., 2024). For example, unobtrusive tactile reminders and allowing users to autonomously review their emotional data are ways to respect their autonomy (Kwok et al., 2024).

Similarly, the "Ocean Pulse" project helps users gain a sense of control and psychological respite by synchronizing breathing with multi-sensory audiovisual and tactile feedback (Kalouaz et al., 2024).

On the technical side, multisensory integration plays a pivotal role in enhancing immersion. The coordinated design of visual, auditory, and tactile stimuli guides attention toward the present experience (Dritsas et al., 2025; Luan & Wang, 2025; Ramtohul & Khedo, 2025; Sirwal et al., 2026). Research shows that gentle, dynamic tactile stimuli can activate emotion-related brain regions (such as the insular cortex), promoting relaxation responses (R. Liu & Kondo, 2025; Tang et al., 2025). Furthermore, some studies have incorporated physiological feedback, enabling the system to detect the user's physiological state in real-time and dynamically adjust interaction parameters and content, thus creating a personalized "perceptual feedback" loop that enhances overall engagement and emotion regulation (Hein et al., 2025; Zhao et al., 2025). This feedback mechanism, informed by psychological frameworks, not only improves user engagement but also strengthens emotion regulation, further supporting the application of kinesthetic mindfulness in emotional management.

However, the existing literature lacks empirical studies on immersive kinesthetic mindfulness interaction guided by psychological theories. Few studies have verified the effects of hand movement-based closed-loop interaction on youth stress relief by combining psychological scales and objective physiological indicators, which is insufficient to support the design and application of related interactive devices.

2.5. Academic contributions of this study

To address the existing gaps in the literature, this study contributes to the field by offering theoretical, design, and empirical advancements. Theoretically, this study integrates the cultural practices of traditional Eastern bead meditation, kinesthetic mindfulness theory, the emotional-touch neural mechanisms, and immersive interaction psychology, thus establishing an emotional regulation framework that links "hand-based kinesthetic engagement - multimodal real-time feedback – mindfulness-rooted awareness." This framework fills a significant gap in the theoretical understanding of the digital transformation of traditional mindfulness tools. In terms of design, the study introduces the "digital prayer beads" concept, combining traditional hand movements in bead meditation rituals with contemporary multimodal interaction technologies. This approach creates a personalized emotion regulation model driven by active kinesthetic engagement, offering a novel paradigm for mental health interaction design aimed at the youth population. Empirically, this study systematically evaluates the effectiveness of the digital prayer beads through the use of psychological scales and physiological indicators, providing a reusable experimental framework and empirical evidence for the assessment of kinesthetic mindfulness digital devices.

3. Design and implementation of the digital prayer bead system

In line with the emotional regulation needs of the youth population discussed in the introduction, the research gap identified through the review of related work, and the primary objectives of this study, this section addresses the research questions RQ1 (How can a kinesthetic mindfulness device be designed for emotional regulation?) and RQ2 (How do the multimodal feedback components in the digital prayer beads work together to regulate the user's emotional state?). It presents the design and implementation process of the digital beads system. To ensure that the design meets the actual needs of the target users (university students and professionals), and to prevent a misalignment between the design and user requirements, we initially conducted a questionnaire survey. This survey gathered insights into the core needs and challenges associated with emotional regulation following stress, providing invaluable empirical data and guiding the direction of the subsequent system design.

3.1. Questionnaire survey

This study aims to explore the emotional states, sources of stress, and regulation needs of young individuals in daily life by conducting surveys among two groups: university students and professionals. Data were collected between September and October 2025. The participant pool consisted of 35 university students (aged 21–30) and 12 professionals, including corporate employees and designers (aged 26–32). Participants were recruited through online channels. Inclusion criteria required that individuals self-report regular emotional stress or anxiety and be capable of independently completing the survey. Exclusion criteria included a history of severe mental illness or recent systematic psychological treatment.

The questionnaire was based on existing validated scales (Demichelis et al., 2024; Keskiner et al., 2024), with minor revisions tailored to the emotional regulation behaviors and haptic interaction preferences relevant to this study. The scales used were previously validated and appropriately adapted to ensure their applicability and accuracy for the target population. The validity of these instruments was confirmed by referenced verified versions, while reliability was ensured through a small-scale pre-survey to assess consistency and stability.

The questionnaire covered three main sections: (1) demographic information and stress source identification; (2) frequency and effectiveness of emotional regulation strategies; and (3) acceptance and expectations for adaptive haptic assistive tools. Data were collected following the principle of anonymity via the online survey platform Questionnaire Star. The study was approved by the Ethics Committee of Hubei University of Technology (Approval No: HBUT20250064).

3.1.1. Feedback from university students

The survey revealed that university students generally experience high levels of stress, with primary sources of pressure being the uncertainty of future opportunities (e.g., employment or further study options, 82.86%), academic workload (e.g., exams, thesis submission, 54.29%), and interpersonal relationships (2.86%). In terms of emotional regulation strategies, the most commonly employed methods were listening to music (74.29%), reading or watching films (57.14%), physical exercise (40%), and talking to friends or family (31.43%). Notably, while students reported declines in work efficiency or mood due to stress, only 17.14% actively sought professional psychological support, with most expressing concerns about psychological counseling. When it comes to new emotional regulation tools, all respondents expressed interest in trying them, with particular emphasis on the portability, privacy of use, and seamless integration with daily routines. They anticipated that the tool would provide a relaxation method that did not rely on electronic screens and could be discreetly used in places such as classrooms or libraries.

3.1.2. Feedback from professionals

Professionals reported diverse sources of stress, with the main concerns being limited career advancement opportunities (80%), blurred boundaries between work and life (73.33%), and excessive workload (53.33%). While many attempted strategies such as exercise and leisure activities, 80% reported that current methods were ineffective or difficult to implement quickly during work breaks. Fifty-three percent of professionals acknowledged that they often ignored or suppressed emotional reactions due to work demands. When considering mindfulness-based tactile feedback devices for emotional regulation, professionals prioritized ease of use and discretion (to avoid disrupting work), the ability to integrate seamlessly into work rhythms (such as providing brief focused breaks), and the scientific and professional integrity of the design. Some respondents emphasized that an ideal tool would help them quickly regain composure between meetings or after high-intensity tasks, without adding additional operational burdens.

3.1.3. Research findings

Based on the questionnaire responses from both university students and professionals, this study identifies the following key findings, which informed adjustments to the subsequent experimental design:

F1: There is a widespread demand for emotional regulation, but existing tools lack sufficient adaptability.

Both groups expressed a significant need for emotional regulation, particularly in high-pressure situations (e.g., exams, meetings), where anxiety severely impacts daily functioning. However, existing regulation strategies (such as self-relaxation techniques, social support, and meditation apps) have proven to be only moderately effective, and professional psychological resources are underutilized due to factors such as high cost and social stigma.

F2: Users have a shared preference for low-intrusiveness, highly portable tools.

Over 80% of respondents emphasized that an ideal emotional regulation tool should be discreet and seamlessly integrated into daily life. For instance, student participants expressed a preference for tools that could be used unobtrusively in classrooms or libraries, while professionals preferred tools that would not cause additional distractions or social embarrassment. This finding suggests that design strategies should avoid explicit "therapy" labels and instead lower the psychological barrier to usage through natural, unobtrusive interactions.

F3: The potential for intervention via motion and tactile feedback is recognized, but context-dependent feedback is essential.

Although motion and tactile interactions are widely viewed as effective in guiding attention, users exhibited divergent preferences for feedback modes: in public settings, participants strongly favored screenless feedback to protect privacy; in private settings, most participants acknowledged the value of multimodal feedback in enhancing immersion. This divergence highlights the difficulty of satisfying diverse needs with a single design, suggesting the need for interaction modes that are adaptable to different contexts, balancing discretion with the depth of experience.

These findings prompted this study to revise the original concept, which combined motion and tactile feedback with audiovisual cues, and to adopt a group experiment approach to more rigorously test the effectiveness of various feedback combinations.

3.1.4. Overall design objectives

Based on the findings outlined above, this study sets the following design objectives:

G1: Evaluate the handheld device for kinetic mindfulness with unimodal and multimodal feedback.

The device focuses on hand motion and tactile interaction, such as sliding beads or controlling a rotating sphere, and is designed to support two feedback conditions. Specifically, Condition A provides only motion-tactile control, with optional adaptive auditory feedback (delivered via headphones), aiming to ensure privacy and subtlety in public settings. In contrast, Condition B adds synchronous visual feedback (e.g., interface animations or lighting effects) to Condition A, further enhancing user immersion. This design effectively addresses the research demand for both "low invasiveness" and "personalized experience," avoiding the limitations of a "universal" approach, and ensures that the emotional regulation needs of diverse users are met.

G2: Function-oriented: brief, high-frequency interventions and closed-loop regulation.

The device should support rapid intervention sessions of 3–5 minutes, where real-time adaptive feedback is driven by the rhythm of hand movements, establishing a closed-loop between "action and feedback." For instance, when the user's movements are steady, the feedback becomes gentle; when the movements are fast, the feedback intensifies. The mechanism integrates physical movement with cognitive anchoring, helping to interrupt repetitive, unproductive thoughts.

G3: Optimize user experience.

Privacy protection is prioritized in the design process, with the device's appearance modeled after everyday items to prevent it from being stigmatized. Furthermore, all feedback modes are designed with privacy in mind, such as limiting visual feedback to the personal screen of the device, ensuring that users' emotional states are not visible in public. Additionally, taking individual differences into account, the device is customizable, allowing users to adjust parameters like feedback intensity and rhythm according to their preferences.

G4: Scientific validation: improve research rigor through experimental group comparisons.

By comparing the effects under various conditions (e.g., anxiety scale scores, heart rate variability data), this study extends beyond merely "validating the effectiveness of multimodal feedback" to explore the adaptive patterns of optimal feedback combinations in different contexts. This shift refines the study's empirical focus, enhancing both its practical applicability and theoretical contribution.

3.2. Conceptual design

This study centers on the design of a digital prayer bead device incorporating adaptive hand-based haptic feedback, intended to ground attention through body movements and facilitate emotional regulation through adaptive feedback. The design draws inspiration from traditional cultural tools, such as Buddhist prayer beads, among others. While these cultural concepts inspire the design, they do not imply formal cross-cultural validation. The conceptual framework is structured around a "perception-processing-feedback loop," emphasizing emotional regulation through interaction between body movements and feedback. The cultural background serves merely as a source of design inspiration.

The device's primary interaction is inspired by the traditional practice of rotating prayer beads. When the user rotates or flicks the beads, the integrated rotational sensor (based on the Hall Effect principle) within the device continuously captures fine hand motion data, such as rotation speed, instantaneous velocity, and motion frequency. These raw data are processed and standardized by an Arduino Uno microcontroller before being transmitted in real-time via serial port to the TouchDesigner visualization programming platform.

Within the TouchDesigner environment, the motion data are converted into control signals that drive audiovisual feedback. A key feature of the system is the establishment of a dynamic correlation between user movements and feedback. For instance, smooth variations in hand rotation speed are mapped to a linear transition in the pitch of environmental sounds or an adjustment in the speed of visual particle flow. This immediate, predictable cause-and-effect relationship is designed to help users focus their attention on their current bodily sensations and the resulting feedback, thereby breaking free from repetitive negative thoughts and entering a state of calm and concentration. This design not only inherits the mind-body integration wisdom of traditional prayer beads for "meditative refinement in motion," but also enhances the immersive and personalized nature of the interaction through digital technology.

3.3. Adaptive feedback logic

The adaptive feedback logic is the core element that enables the device to offer personalized emotional regulation. The objective is to dynamically adjust audiovisual feedback parameters based on the user's real-time behavior characteristics to either reflect the user's current emotional state or guide it toward a desired state. The adaptive feedback process is shown in Figure 2.

In the state-following mode, the system's feedback aims to mirror the user's current state. By monitoring the behavior data stream from the Arduino, the system calculates key features such as rotation speed, speed variance (stability), and motion abruptness. These features serve as control variables, which are then functionally linked to target audiovisual parameters (e.g., the speed and amplitude of visual particle diffusion, and the fundamental frequency and rhythm of sound). For example, when the system detects that the user's actions are fast and unstable, the feedback adjusts to higher frequency, faster-paced audio and dynamic visual effects. This feedback design allows the user to perceive their emotional state and, by consciously slowing down their hand movements, actively participate in emotional regulation, ultimately guiding the feedback toward a more tranquil state, thus helping the user return to a more stable emotional condition (Cheung et al., 2025).

The state-following mode not only passively responds to user actions but also guides users to adjust their emotions through real-time feedback regulation. As users engage with the device, they actively perceive changes in the feedback and adjust their motion rhythm to influence the feedback, thereby achieving emotional regulation. Through rhythmic hand movements, users naturally trigger a relaxation response, rather than simply receiving passive instructions. This mode emphasizes the continuous nature of hand movements, making the movement itself the central element of the emotional regulation process.

3.4. Audiovisual feedback implementation

3.4.1. Visual feedback system construction

The visual feedback system is developed within TouchDesigner, primarily employing its real-time particle system and video feedback loop technology to generate immersive visual scenes (J. Huang et al.,

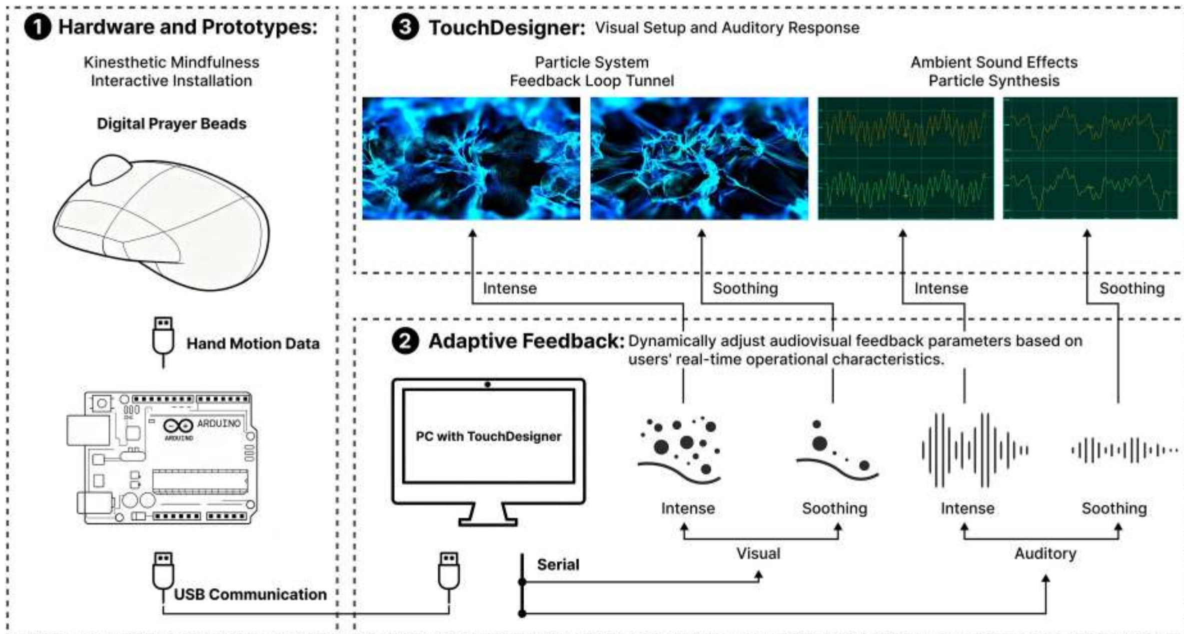

Figure 2. Adaptive feedback process: (1) Hardware and prototype; (2) Adaptive PC-based feedback: Auditory and visual; (3) Specific auditory and visual feedback presentation.

2025; Qiu & Zhang, 2025). The core visual scene typically consists of multiple particle generators (Particle SOP/CHOP), with attributes such as emission rate, lifespan, movement speed, and color being directly correlated with the rotational speed data input from Arduino.

To achieve dynamic and diverse visual effects, a feedback loop technique with tunnel effect is utilized. Specifically, the particle rendering output image is fed back into the start of the rendering pipeline through Delay TOP or Feedback TOP components and is superimposed, blended, or distorted with the new particle visuals (Feedback TOP – TouchDesigner Documentation, 2025). By incorporating components such as Lens Distort TOP and Noise TOP, and dynamically modulating their parameters, fluid-like motion, infinite extension, or organic distortion visual forms can be produced. For example, a slowly rotating element corresponds to a smooth, expanding spiral movement of particles, while rapid rotations trigger an explosive diffusion of particles. Additionally, color correction components like Level TOP, HSV Adjust TOP, and Luma Blur TOP allow real-time adjustments to the overall tone, brightness, and saturation of the image, creating different visual atmospheres such as calm, active, or dreamlike, depending on the interactive context.

3.4.2. Auditory feedback mechanism

Auditory feedback is realized through the built-in audio synthesis (CHOPs) and sample playback functions within TouchDesigner. The system maps hand movement data to control low-frequency oscillator (LFO) signals for audio synthesis or directly adjusts the playback parameters of audio files. A typical approach involves synthesizing environmental sound stimuli (Serrano & Cartwright, 2023). For instance, rotational speed is mapped to the frequency and amplitude of a sine wave or noise generator, producing sounds such as wind, flowing water, or ethereal background noise. As the rotational speed increases, the pitch rises and the volume intensifies, creating a sense of tension; when the speed decreases, the pitch drops and the volume diminishes, inducing a calming effect. Another method is granular synthesis (Roads et al., 2021), which decomposes a short audio sample into minute grains and adjusts the timbre and rhythm by controlling the speed, density, and sequence of the grains, with these parameters also driven by hand movements. This study primarily adopts the first method.

To ensure the cohesion of the audiovisual experience, the system utilizes the Analysis CHOP component to analyze the generated audio signal in real-time (e.g., volume or spectral centroid) and feeds it back into the visual particle system. For instance, the audio amplitude can be used to drive particle size, thereby achieving audio-visual alignment and enhancing the overall sense of multimodal feedback and immersion.

3.5. Hardware key points and prototype

3.5.1. Design philosophy

The primary design philosophy for the hardware is non-intrusiveness and seamless integration. The device's appearance draws inspiration from traditional prayer beads or cultural ornaments, avoiding the visual impression of "medical equipment" or "experimental devices" to reduce the psychological barrier to usage and encourage natural incorporation into everyday life contexts, as illustrated in Figure 3(a,c). The core of its interaction is to preserve the purity of tactile feedback during user behavior, meaning that users do not need to learn complex operations and can interact with the system through intuitive hand movements such as rotation and flicking.

3.5.2. Appearance design

The prototype device is designed with a handheld form that conforms to the contours of the hand, featuring bead-like spheres that can be flicked with the thumb, as shown in Figure 3(c). The casing is made from a dermal-friendly material (thermoresponsive resin), and its dimensions are carefully crafted to approximately 12.6 cm (length) × 8.9 cm (width) × 7.6 cm (height), ensuring comfort during extended use. The key sensor unit is seamlessly integrated into the device, achieving a harmonious balance between function and form. The overall design emphasizes simplicity and subtlety, allowing it to be used in public settings such as classrooms or offices without attracting undue attention, while addressing the user's needs for privacy and convenience.

3.5.3. Function implementation

The core sensing mechanism employs Hall effect sensors (Ying & Hall, 2021). These sensors consist of a plastic valve body, a flow rotor assembly, and a Hall sensor chip, as shown in Figure 3(a,b). The rotor assembly contains permanent magnets, and when the user rotates the bead, causing the rotor to spin, the Hall sensor detects changes in the magnetic field and outputs corresponding pulse signals. The Arduino Uno microcontroller receives these pulse signals, calculates the pulse frequency to accurately determine rotation speed, and counts the pulses to accumulate rotation angles. After initial processing, this data is reliably transmitted to the TouchDesigner software running on a PC via serial communication (UART), enabling haptic interaction and supporting the emotional regulation design philosophy.

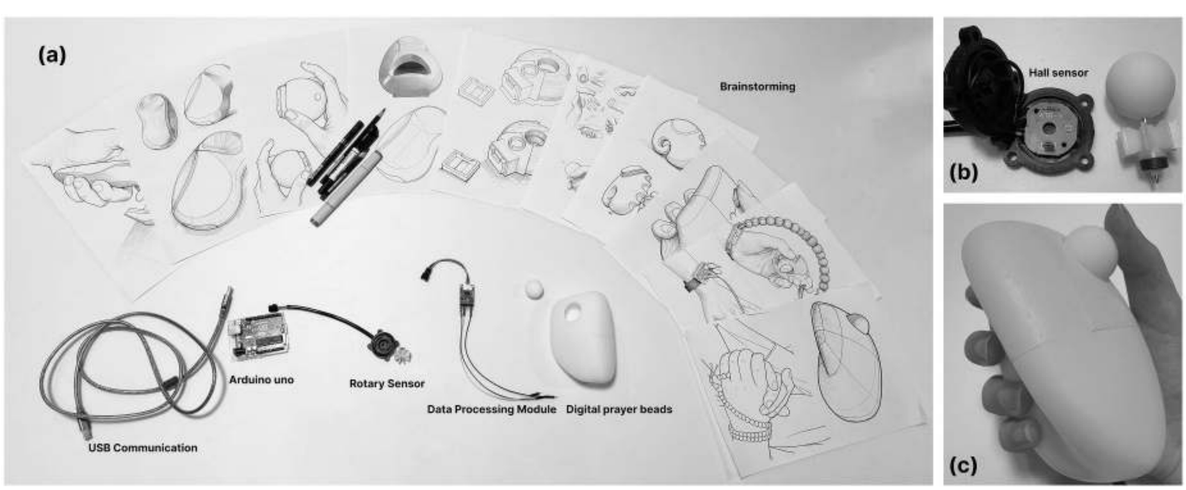

Figure 3. Hardware design and prototype implementation: (a) brainstorming and specific hardware prototype components; (b) Hall-effect-based rotation sensor; (c) digital prayer beads prototype.

3.6. Pilot study

To systematically assess the feasibility of emotional regulation and the stability of the technical solution proposed in this study, we conducted a pilot study prior to the main experiment. The primary goal was not to validate the intervention's effectiveness but to test the experimental process, equipment reliability, data collection quality, and participant experience on a smaller scale. This allowed us to identify potential issues and inform the optimization of the formal research protocol.

The pilot study recruited 3 participants: 2 university students and 1 working professional, all of whom signed informed consent forms. The study took place in a controlled laboratory setting, with each participant experiencing the entire procedure, from stress induction and device interaction to post-assessment. Regarding technical reliability, we focused on monitoring the continuity and accuracy of sensor data collection, the stability and latency of data transmission between the Arduino and the TouchDesigner platform, and the synchronization and lag of audiovisual feedback. The flow of the procedure was evaluated by observing the naturalness and time efficiency of transitions between experimental phases, such as instruction delivery, task switching, and device operation. Subjective feedback was collected through semi-structured interviews and simple questionnaires, with an emphasis on participants' initial impressions of device comfort, interaction logic, and feedback mode acceptability.

3.6.1. Results and improvements

The data collected during the pilot study provided valuable insights for the design and execution of the formal experiment. Based on pilot observations and interview feedback, several key improvements were made to the experimental protocol:

1. To address the slight delay in the rotational feedback sound during high-speed operation, we optimized the data parsing nodes in TouchDesigner to improve real-time responsiveness.
2. To reduce cognitive load for first-time users, we refined the instructional language and introduced a brief practice session.
3. Based on ergonomic feedback, the handle of the hardware prototype was adjusted to enhance comfort during prolonged use.

In conclusion, this pilot study effectively evaluated the research process and technical system, providing crucial practical insights and optimization recommendations that will ensure the efficient and reliable execution of the formal study, thereby enhancing the scientific validity and persuasiveness of the research findings.

4. Methodology

To answer research questions RQ3: What are the subjective experiences, usability, and stress-relief effects of users when interacting with the digital prayer beads device? RQ4: How does the digital prayer beads device influence users' attention focus and present-moment awareness? This study investigates the impact of a hand-based tactile interaction digital bead device on users' anxiety reduction and emotional experiences, comparing the effects of different feedback conditions. The mindfulness experiment is based on tactile feedback, specifically comparing two feedback conditions: auditory feedback and multimodal feedback combining both auditory and visual stimuli. A nature restoration group is included as the control group. The primary aim of this study is to assess the feasibility of the device and evaluate its effectiveness in reducing stress under various feedback conditions. The following hypotheses are proposed:

H1: After engaging in a tactile mindfulness intervention with the "digital bead" device, users' anxiety levels will significantly decrease, and their physiological arousal levels will notably reduce.

H2: Among tactile interaction-based feedback conditions, multimodal feedback (auditory + visual) will be more effective in lowering users' anxiety levels than auditory-only feedback.

H3: Multimodal feedback will lead to more positive emotional experiences for users compared to auditory-only feedback, and multisensory stimulation will enhance users' sense of immersion and control.

4.1. Participants

This study recruited 30 adult participants through a combination of online public recruitment and offline referrals. Among them, 25 were university students (aged 21–25 years) and 5 were working professionals (aged 25–30 years), all of whom met the following inclusion and exclusion criteria to ensure the validity of the sample and the ethical safety of the experiment.

- Inclusion criteria: (1) Self-reported experience of regular emotional stress or anxiety; (2) Ability to independently complete experimental tasks.
- Exclusion criteria: (1) History of severe mental illness or recent systematic psychological treatment; (2) Significant cognitive or motor dysfunction that would interfere with experimental procedures.

The sample included more university students than working professionals, which was primarily based on the feasibility of recruitment and the study's stage. As an exploratory study, its core aim was to preliminarily assess the feasibility and effectiveness of the digital prayer bead intervention. University students were easier to recruit centrally through campus channels, with flexible schedules, allowing for stable sampling. The inclusion of working professionals aimed to enhance sample diversity and provide an initial test of the study's external validity.

All participants signed written informed consent forms, and their experimental data were stored with anonymized identifiers to protect privacy. This study was approved by the Ethics Committee of Hubei University of Technology (Approval No: HBUT20250064) and adhered strictly to academic ethical standards, ensuring the experiment's compliance and safety.

4.2. Experimental design

This study employs a one-factor between-group design to compare the impact of different feedback types on emotion regulation. The independent variable, "feedback type," consists of the following three levels:

Natural Recovery (NR): Serving as the control group, participants undergo no intervention after stress induction, allowing them to recover naturally without using any devices or performing specific actions.

Auditory Feedback (AF): Participants use a digital mindfulness device, such as a prayer beads system, that provides adaptive auditory feedback based on hand movements.

Audiovisual Feedback (VAF): In addition to the auditory feedback in AF, visual feedback is presented synchronously to enhance the experience.

To ensure that the effects on emotion regulation are specifically due to the kinesthetic interaction, we implemented the following control measures:

- Control and Experimental Group Design: The NR group acts as the control, where no intervention is applied after stress induction, and only natural recovery occurs. This design helps eliminate any potential influences from interventions, allowing us to test whether kinesthetic interaction can independently promote emotion regulation and stress relief. In contrast, both the AF and VAF groups engage in kinesthetic interaction through auditory and audiovisual feedback, respectively, ensuring that observed differences are due solely to the unique effects of kinesthetic interaction.
- Random Assignment and Group Balancing: Using the RAND() function in Microsoft Excel to generate uniformly distributed random numbers, participants ($n = 30$) were randomly assigned to one of three experimental groups. Each participant was given a unique identifier, and the random numbers were used to rank participants, dividing them evenly into three groups, each consisting of 10 individuals. Each group was exposed to only one feedback condition. To prevent potential expectation effects, participants were blinded to their group assignments. This approach minimizes potential biases due to inter-group differences, ensuring that any observed effect differences are

attributable to the feedback condition. Baseline data, including participants' age, gender, and STAI-S scores, were collected before the experiment. Continuous variables were analyzed using one-way ANOVA, while categorical variables were compared across groups using the chi-square test. No statistically significant baseline differences were found between the three groups, ensuring consistency for subsequent experimental analysis.

- Expectation Effect Control: Prior to the experiment, participants are clearly briefed on the study's purpose and tasks to minimize the influence of expectation effects on the results. All participants receive standardized instructions to avoid any psychological placebo effect based on their expectations.
- Standardization and Consistency in Procedures: To minimize novelty effects and task-related interference, all other experimental conditions, such as the environment (lighting, noise), stress induction procedure, interaction tasks (with no action for the NR group), experimental duration, and physiological data collection processes, are kept consistent across all conditions.

Through these control measures, we ensure that any observed effects on emotion regulation can be attributed to the kinesthetic interaction itself, rather than to novelty effects, expectation effects, or merely performing a calming task.

4.3. Experimental procedure

This experiment followed a standardized, sequential procedure, where all participants adhered to the same steps, with the only difference being the type of feedback provided during the intervention phase between groups. The overall process consisted of: experimental briefing and preparation, baseline calibration, pretest, stress induction, interactive intervention, post-test, and interview, as detailed below.

4.3.1. Experimental preparation and baseline calibration

Prior to the formal experiment, the researchers thoroughly explained the experiment's objectives, procedure, potential risks, and privacy protection measures to the participants. Once the participants confirmed their understanding, they signed a written informed consent form.

Next, the experimental environment was standardized, and calibration and synchronization tests were conducted for both the physiological data collection equipment and the interactive system of the digital prayer beads. This ensured the accuracy of the data timestamps and the stability of the equipment.

To establish a stable physiological baseline, participants were instructed to perform the 4-7-8 breathing exercise, which helps stabilize both emotional and physiological states (Aktaş & İlgin, 2023). The overall experimental procedure is illustrated in Figure 4.

4.3.2. Pretest phase

After baseline stabilization, all participants completed the State-Trait Anxiety Inventory (STAI) (Ding et al., 2022; Valera et al., 2024) to measure their initial anxiety levels before the stress induction.

Specifically, the STAI-S state anxiety scale was used to assess immediate anxiety levels and served as the primary measure for comparison before and after the intervention. The STAI-T trait anxiety scale recorded the participant's long-term anxiety tendencies, providing a basis for subsequent individual difference analysis.

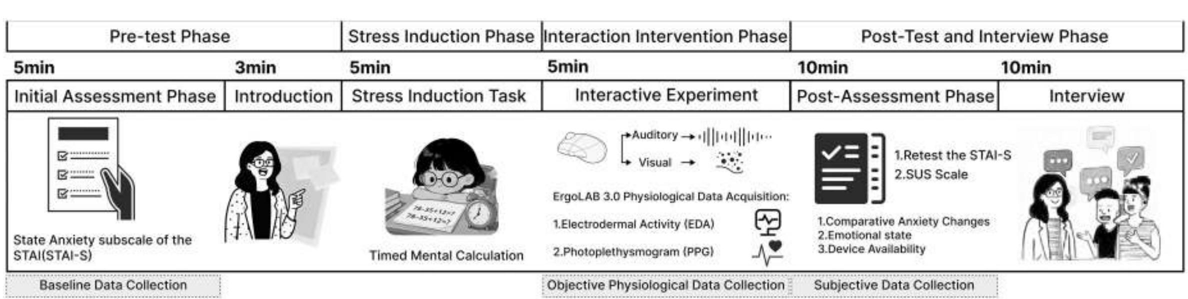

Figure 4. Experimental procedure.

4.3.3. Stress induction phase

For stress induction, the modified Trier Social Stress Test (TSST) mental arithmetic task was used as the standardized stressor (Gunnar et al., 2021).

Participants completed a 5-minute standardized mental arithmetic task, with all questions selected from a pretested question bank. The task's structure, difficulty level, time constraints, and cognitive load were kept consistent to ensure uniform stress induction across groups.

Time pressure and performance feedback were applied throughout the task to effectively trigger both psychological and physiological stress responses.

4.3.4. Interaction intervention phase

Following the stress-inducing task, participants transitioned immediately into a 5-minute emotion regulation intervention phase. During this phase, they performed the tasks based on their randomly assigned condition:

- NR: Participants sat quietly to rest, without engaging with the device or performing any additional tasks.
- AF: Participants interacted with the digital prayer bead system and received adaptive auditory feedback.
- VAF: Participants interacted with the digital prayer bead system and received synchronized adaptive audiovisual feedback.
- The experimental procedure remained consistent across all groups, with the only independent variable being the type of feedback provided. During the intervention, physiological data, including skin conductance response (EDA) and heart rate variability (HRV), were continuously and synchronously recorded.

4.3.5. Post-test and interview phase

Immediately after the intervention, a post-assessment was conducted:

- Quantitative Post-Test: Participants completed the STAI-S scale again, allowing for a comparison of pre- and post-intervention scores to assess anxiety levels. Additionally, the System Usability Scale (SUS) was administered to evaluate the overall usability and user experience of the device (Alshamari & Althobaiti, 2024).
- Qualitative Interview: A semi-structured interview lasting approximately 10 min was conducted in a private setting, covering topics such as participants' usage experiences, emotional changes, comparisons with other relaxation techniques, and suggestions for improvement. With participants' consent, the interviews were recorded and transcribed for subsequent thematic analysis (Kang et al., 2025; Zhang & Li, 2025).

4.4. Data collection

In this study, subjective psychological and objective physiological data were collected at corresponding experimental time points. The data collection details and timing are as follows:

4.4.1. Subjective data

- Pretest: STAI-S, STAI-T;
- Post-test: STAI-S, SUS;
- Qualitative Data: Transcripts from semi-structured interviews conducted after the intervention.

Following data entry, completeness checks were performed. Samples with more than 20% missing data were excluded, and scale scores were standardized for ease of inter-group comparison.

4.4.2. Physiological data

Physiological indicators were continuously recorded during the intervention phase:

- EDA: Indicates sympathetic nervous system activation and emotional arousal;
- HRV: Reflects autonomic nervous system balance and emotional regulation capacity.

All physiological signals were synchronized with experimental events (stress onset, intervention onset, feedback trigger) to ensure precise subsequent analysis.

4.5. Data processing and statistical analysis

4.5.1. Subjective data processing

The valid scale data were reverse scored and the total score was calculated, followed by normalization to eliminate dimensional differences. This was done for analyzing anxiety changes, usability evaluation, and the relationship with individual differences.

4.5.2. Physiological data processing

The raw signals of EDA and HRV were preprocessed as follows:

- EDA: Initially, systematic artifact detection and noise removal are performed. Gaussian filtering is applied for noise filtering, followed by decomposition into phase and tension components. Key quantifiable features, such as signal amplitude and fluctuation frequency, are then extracted (Amin & Faghih, 2022).
- HRV: Abnormal waveforms resulting from body movements or equipment interference are strictly identified and removed. Errors in R-wave detection and abnormal pulse signals are corrected, resulting in a clean and valid R-R interval sequence. Based on this, feature calculation and analysis are carried out in both time and frequency dimensions (Damoun et al., 2024).

All physiological features were extracted in segments according to experimental stages and precisely synchronized with key time points.

4.5.3. Statistical analysis

Statistical analysis was performed using IBM SPSS 27, with a significance level of $\alpha = 0.05$.

- Group Comparisons: If the data is normally distributed, one-way analysis of variance (ANOVA) was used, otherwise the Kruskal-Wallis nonparametric test was applied.
- Effect Size: η^2 was calculated to assess the practical significance of the intervention effects.
- Subsequent Analysis: The relationship between scales and physiological data was tested to verify the research hypotheses and analyze the impact of individual differences.

4.6. Materials and equipment

To ensure the accuracy, consistency, and repeatability of experimental data collection, all interaction experiments in this study were conducted in the dedicated laboratory of the User Research and Experience Design Institute at Hubei University of Technology. The laboratory environment was standardized to minimize the interference of external variables on participants' physiological indicators and emotional states. The specific experimental materials are shown in Figure 5.

4.6.1. Experimental environment

The experimental environment was fully standardized to eliminate external influences and ensure that all participants had a consistent experience. This setup ensured the stability of both physiological data and emotional states. The specific control parameters are as follows:

- Noise Control: The laboratory is designed to be soundproof, maintaining a background noise level below 45 dB to prevent interference with the participants' emotions and attention.
- Lighting Control: Adjustable lighting is used, with the intensity set between 300–500 lux, a comfortable range for the participants. The lighting conditions at all experimental stations are uniform, ensuring that lighting differences do not affect participants' emotional states.

Figure 5. Experimental materials: (a) specific experimental apparatus; (b) experimental setting and participants.

- Temperature Control: The room temperature is maintained at $20 \pm 1\,^{\circ}\mathrm{C}$ using a central air conditioning system, ensuring participant comfort and reducing the impact of temperature on physiological measurements.
- Layout Design: The layout consists of independent partitions (width $\geq$ 2.5 meters, length $\geq$ 6.2 meters) to provide participants with isolated workspaces free from disturbances. One-way observation mirrors are installed, allowing researchers to observe and record behavior without intrusion (as shown in Figure 5(b)).
- Operational Standards: All experimental procedures are conducted under the guidance of a single researcher to ensure consistency across all stages, which enhances the reproducibility of the experiment.

4.6.2. Equipment configuration

The experimental setup consists of three categories of equipment based on function: core interactive devices, physiological monitoring devices, and auxiliary devices. All equipment underwent standardized calibration prior to the experiment to ensure stable performance and precise data collection. The configuration is as follows (refer to Figure 5(a)):

- Core Interactive Devices: A handheld digital mindfulness device, integrated with a Hall effect sensor to detect hand rotation movements. This device connects to Arduino and TouchDesigner, enabling real-time audiovisual feedback programming to ensure timely interaction responses. For detailed specifications, refer to Section 3.5.
- Physiological Monitoring Devices: The ErgoLAB 3.0 human-machine environment synchronization platform, along with accompanying sensors, uses wireless transmission to minimize participant interference:
  - EDA Skin Conductance Sensor (sampling rate 64Hz): Measures skin electrical activity to assess sympathetic nervous system arousal.
  - PPG Wireless Blood Volume Pulse Sensor (sampling rate 64Hz): Monitors heart rate (HR) and HRV, reflecting the balance between sympathetic and parasympathetic nervous system activity.
- Auxiliary Devices:
  - iPad 9: Utilized for collecting data through questionnaires (STAI, SUS), offering easy touchscreen operation and enabling real-time data uploads to the data collection platform.
  - Lenovo Y9000P 2023 Laptop: Generates adaptive audiovisual feedback for the interactive system.
  - BOSE QuietComfort 45 Noise-Canceling Headphones: Deliver auditory feedback while eliminating environmental noise interference to maintain feedback purity.
  - 2K 240Hz Monitor: Displays visual feedback, ensuring smooth imagery and enhancing participant immersion.
- Hygiene Assurance: All sensors and equipment are thoroughly disinfected with tissues and alcohol before and after each use to ensure the hygiene and safety of participants.

5. Results

This study evaluated the effects of different feedback types on participants' emotional regulation and stress relief through various physiological indicators. By comparing the differences between the NR, AF, and VAF groups, we examined the impact of feedback methods on autonomic nervous system regulation, emotional stability, and stress alleviation.

5.1. STAI-S results

To ensure consistency across different scoring dimensions and make the results comparable, the study applied a min-max normalization technique. This transformed all scores into a standardized range from 1 to 100, eliminating dimensional differences between the various scoring categories.

Based on the normalized scores, the STAI-S scores for each group before and after the experiment were as follows: The NR group had pre- and post-experiment scores of 67.33 and 51.67, respectively; the AF group had scores of 62 and 17.33; and the VAF group had scores of 60 and 14.67, as shown in Figure 6. To examine potential differences in STAI-S scores among the groups, a normality test was first conducted. The results indicated that some data did not follow a normal distribution, leading to the use of the Kruskal-Wallis non-parametric test for further analysis. The results of the non-parametric test revealed no significant differences in STAI-S scores between the three groups prior to the experiment ($p = 0.329$), suggesting that the initial stress levels of all groups were similar and confirming the validity of the experimental procedure. The inter-subject effect test ($F(2,\ 27)\ =\ 1.632$, $p = 0.214$, Partial $\eta^2\ =$ 0.108) further supported that there were no statistically significant differences before the experiment.

After the intervention, significant differences were observed in the STAI-S scores between the three groups ($p < 0.001$). Further analysis of inter-subject effects ($F(2,\ 27)\ =\ 61.408$, $p < 0.001$, Partial $\eta^2\ =$ 0.820) indicated a strong main effect of feedback type on STAI-S scores, explaining 82.0% of the score variability. Specifically, post-experiment STAI-S scores followed the pattern: NR > AF > VAF, with significant statistical differences between all groups. Notably, the VAF group showed the lowest STAI-S scores, highlighting the significant impact of audiovisual feedback in reducing state anxiety.

These findings suggest that the use of "digital beads" for mindfulness feedback can effectively reduce state anxiety in participants, particularly under the combined audiovisual feedback condition. This indicates the positive influence of multisensory feedback in emotional regulation and stress relief.

5.2. EDA results

EDA is utilized to assess participants' psychophysiological responses, including SCL (skin conductance level) and SCR (skin conductance response). SCL represents an individual's baseline skin conductance level, reflecting tonic responses typically related to emotional and physiological baseline states. In contrast, SCR measures the transient response to external stimuli, corresponding to phasic responses, and is generally associated with emotional fluctuations or stress responses (Hamilton et al., 2024; Ney et al., 2024).

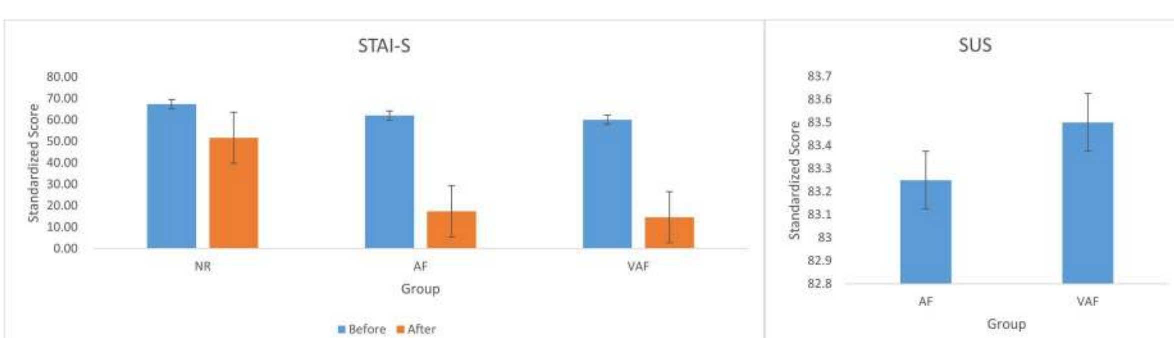

Figure 6. Subjective data results: STAI-S: Standardized scores before and after the experiment; SUS: Standardized system usability scores for the adaptive mindfulness feedback groups.

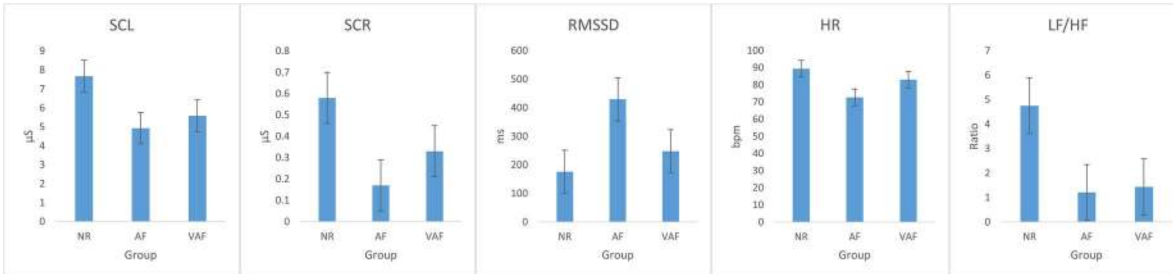

Figure 7. Physiological data results: EDA: SCL and SCR; HRV: RMSSD, HR, and LF/HF.

5.2.1. SCL

The experimental results revealed the following mean SCL values for the three groups: NR group SCL $= 7.68\ \mu$S, AF group SCL $= 4.93\ \mu$S, and VAF group SCL $= 5.58\ \mu$S, as illustrated in Figure 7. Specifically, the SCL values in the NR group were significantly higher than those in the VAF and AF groups, suggesting that participants in the two groups using the "digital prayer beads" for adaptive mindfulness feedback exhibited lower overall arousal and more relaxed physiological states compared to the natural recovery group, potentially making them more effective in alleviating stress. However, the SCL in the VAF group was higher than that in the AF group, likely due to the cognitive load or increased attention elicited by the visual stimuli, which may have provided some additional benefit for stress relief.

To further investigate the differences in SCL, a Kruskal-Wallis non-parametric test was applied due to the non-normal distribution of the data. The results revealed statistically significant differences in SCL across the feedback groups ($p = 0.033$). Moreover, an inter-subjects effect test ($F(2,\ 27) = 4.07$, $p = 0.029$, Partial $\eta^2 = 0.232$) indicated that the type of feedback had a substantial effect on SCL, explaining 23.2% of the variance, making it a key factor influencing SCL.

5.2.2. SCR

Regarding the SCR results, the mean SCR for the NR group was 0.58 μS, for the AF group was 0.17 μS, and for the VAF group was 0.33 μS, following the order NR > VAF > AF, as shown in Figure 7. These findings indicate that the two groups using "digital prayer beads" for adaptive mindfulness feedback exhibited better emotional stability and higher stress response thresholds compared to the natural recovery group, suggesting that adaptive feedback facilitates emotional regulation and stress relief. The SCR in the VAF group was higher than in the AF group, which may be attributed to the increased cognitive load or visual processing demands from the visual feedback, such as smooth eye movements or pupil tracking, which could have enhanced stress responses.

Normality testing showed that SCR followed a normal distribution, and subsequent one-way ANOVA indicated significant differences in SCR between the feedback groups ($p < 0.001$). Further inter-subjects effect tests ($F(2,\ 27) = 10.01$, $p < 0.001$, Partial $\eta^2 = 0.426$) confirmed that feedback type had a significant main effect on SCR, with a large effect size.

5.3. HRV results

HRV is a crucial physiological indicator for assessing the function of the autonomic nervous system. By analyzing variations in the intervals between successive heartbeats (i.e., R-R intervals), HRV helps evaluate an individual's self-regulation and recovery capacity. HRV analysis includes time-domain measures (e.g., RMSSD) and frequency-domain measures (e.g., LF/HF ratio), offering insights into the balance between sympathetic and parasympathetic nervous system activity (Ali et al., 2021; Gullett et al., 2023).

In time-domain analysis, RMSSD (root mean square of successive differences) is a widely used indicator, reflecting the variability of heart inter-beat intervals, primarily regulated by parasympathetic nervous activity (Schöneburg et al., 2024). Higher RMSSD values generally indicate stronger parasympathetic activity, which corresponds to better self-regulation and recovery capacity, typically linked to relaxation, rest, and improved psychophysiological states. Additionally, HR is a fundamental measure in HRV analysis, reflecting the number of heartbeats per unit time (Quigley et al., 2024). HR is associated with autonomic nervous system balance: elevated HR often correlates with sympathetic activation and stress, while lower HR is associated with enhanced parasympathetic activity, relaxation, and recovery.

The LF/HF ratio, a key metric in frequency-domain analysis, provides a deeper understanding of the autonomic balance by analyzing the frequency spectrum of HRV. The LF (low-frequency) band reflects both sympathetic and parasympathetic activity, while the HF (high-frequency) band is predominantly associated with parasympathetic activity (Pham et al., 2021). The LF/HF ratio is used to assess the relative activity of the sympathetic and parasympathetic systems. Higher LF/HF ratios suggest sympathetic dominance, often associated with stress and anxiety, whereas lower ratios indicate parasympathetic dominance, signaling relaxation and recovery.

5.3.1. HRV time-domain analysis: RMSSD and HR results

The time-domain analysis revealed the following mean RMSSD values: NR = 176.14 ms, AF = 429.94 ms, VAF = 247.62 ms, as shown in Figure 7. Notably, the AF and VAF groups exhibited higher RMSSD values compared to the NR group, indicating stronger parasympathetic activity and suggesting enhanced self-regulation and recovery in these groups. The VAF group's RMSSD was lower than the AF group, likely due to the additional cognitive load induced by visual feedback. Kruskal-Wallis non-parametric testing indicated no statistically significant difference in RMSSD ($p = 0.141$). However, inter-subject effect analysis revealed that feedback type approached statistical significance ($F(2, 27) = 2.98$, $p = 0.068$, Partial $\eta^2 = 0.181$), indicating that feedback type influenced RMSSD, though not to a statistically significant degree.

Regarding HR, the NR group showed a significantly higher HR (89.50 bpm) compared to the AF group (72.80 bpm) and VAF group (83.10 bpm), as shown in Figure 7. This suggests that the adaptive feedback groups were more relaxed, experiencing more effective stress relief than the natural recovery group. Kruskal-Wallis testing confirmed significant differences in HR ($p < 0.001$), and inter-subject effect analysis ($F(2, 27) = 10.95$, $p < 0.001$, Partial $\eta^2 = 0.448$) further validated the substantial impact of feedback type on HR.

5.3.2. HRV frequency-domain analysis: LF/HF ratio results

The LF/HF ratio in the NR group (4.75) was significantly higher than in both the AF group (1.21) and the VAF group (1.44), as shown in Figure 7, suggesting parasympathetic dominance in the feedback groups, which were associated with higher relaxation and stress relief. The slightly higher LF/HF ratio in the VAF group compared to the AF group may reflect different cognitive loads due to multisensory feedback, affecting the autonomic balance. Kruskal-Wallis testing showed that differences in the LF/HF ratio were statistically significant ($p = 0.037$). Inter-subject analysis ($F(2, 27) = 3.59$, $p = 0.041$, Partial $\eta^2 = 0.210$) further corroborated the effect of feedback type on LF/HF balance, highlighting its role in stress regulation and emotional stability.

These physiological findings suggest that different feedback types (NR, AF, VAF) significantly influence emotional regulation and stress relief. Kinesthetic mindfulness training with the "digital prayer beads" device enhanced parasympathetic activity, alleviated stress, and regulated emotions more effectively than natural recovery. Moreover, feedback type had a marked impact on physiological metrics, such as EDA (SCL and SCR), HRV (RMSSD, HR, and LF/HF ratio), underscoring the potential of body movement-based kinesthetic interaction in psychological health interventions.

5.4. SUS results

SUS was used to evaluate the usability and user experience of the "digital prayer beads" device. The NR group, which did not use any device, was excluded from the SUS evaluation. For the AF and VAF groups, after standardization, the SUS scores were 83.25 and 83.5, respectively, as shown in Figure 6. These high and similar scores indicate that participants rated the device positively in terms of usability and user experience. These results suggest that the device performed well in terms of operability and feedback adaptability, resulting in a satisfactory user experience.

5.5. Interview results

To better understand the participants' user experiences, this study conducted semi-structured interviews following the "Digital Prayer Beads" device's kinesthetic mindfulness training experiment. The goal was to supplement quantitative data with qualitative insights and explore the application effects of the device. The interview data were analyzed using thematic analysis. To ensure reliability and consistency, two independent researchers performed the coding process. They first reviewed all interview records multiple times to identify potential themes and patterns, then labeled significant information relevant to the research questions and coded it independently. Regular discussions were held to ensure coding consistency and accuracy. To further assess the reliability of the results, the research team calculated coding consistency using Cohen's Kappa coefficient, resolving any discrepancies through negotiation (Garcia et al., 2025).

Six key themes were identified from the interview data: kinesthetic mindfulness training experience, tactile feedback of the device, perception of adaptive feedback, user strategies influenced by individual differences, the impact of cultural background on interaction preferences, and reflections on stress induction and regulation effects. These analyses provided a deeper qualitative perspective on participants' multidimensional experiences in kinesthetic mindfulness training, supporting the experimental results.

5.5.1. Experience with kinesthetic mindfulness

During the interviews, most participants reported that kinesthetic mindfulness training effectively helped them focus and enter a meditative state. The combination of visual and auditory feedback was especially noted as key to maintaining focus. A participant from the VAF group commented: "When the visual and auditory feedback combined, I could feel my mood gradually calm, and my attention stayed focused." In contrast, participants from the NR group felt that, without any external feedback, it was harder to stay focused, and the mindfulness effect was limited. One participant from the NR group reflected: "Without external feedback, my thoughts easily wandered. Although I knew I should relax, I felt it was hard to really relax."

5.5.2. Reaction to the device's tactile feedback

Participants' reactions to the tactile feedback of the "Digital Prayer Beads" device were mixed. Some participants enjoyed the texture and tactile feedback, believing it enhanced their mindfulness connection. A participant from the VAF group said: "The feeling of rolling my fingers on the beads is very comfortable, like performing a meditative ritual, which helps me stay focused." However, some participants reported discomfort with the device's hardness and tactile sensation, especially after prolonged use, which led to finger fatigue. A participant from the AF group noted: "The device's hardness made my fingers slightly uncomfortable, and after long use, my palms felt sore."

5.5.3. Perception of adaptive feedback

Most participants felt that the combination of visual and auditory feedback provided comprehensive feedback during kinesthetic mindfulness training, helping them clearly perceive their emotional and physiological states. A participant from the VAF group said: "When visual and auditory feedback came together, it helped me clearly know if I was in the right relaxed state, the feedback was very intuitive." However, some participants found visual feedback sometimes burdensome, especially when it was complex. A participant from the VAF group remarked: "Sometimes the visual feedback distracted me, and I felt unable to fully focus on my bodily sensations."

5.5.4. Individual differences in usage strategies

The interview results revealed that individual differences significantly impacted participants' usage strategies. Some participants preferred visual feedback, believing it clearly reflected their meditation progress. One participant stated: “The visual feedback helps me clearly understand my status.” Others preferred simpler auditory feedback, finding it direct and effective. A participant said: “Auditory feedback is simple and direct, it doesn’t overwhelm me.” Additionally, some participants adjusted their strategies based on need, like slowing movements to enhance tactile feedback or reducing frequency to avoid cognitive overload. Although this study did not explore personality traits specifically, it can be inferred that extroverted or stimulus-seeking individuals may prefer multimodal feedback, while introverted or cautious individuals may prefer simpler feedback.

5.5.5. Cultural background and interaction preferences

All participants in this study were from China and generally preferred subtle, reserved feedback methods, favoring interactions that were simple and had a strong meditative experience. One participant shared: “I am used to traditional meditation methods, and the tactile feedback from this device feels ritualistic, helping me enter a meditative state.” During the experiment, participants generally believed that tactile feedback offered better guidance and aligned with their meditation habits and psychological needs.

5.5.6. Reflections on stress induction and regulation effects

Most participants reported feeling more relaxed during meditation with the “Digital Prayer Beads” device, especially with adaptive feedback. Participants from the VAF group generally believed that combining visual and auditory feedback significantly reduced their stress and anxiety. A participant from the VAF group said: “During the process, I felt my stress gradually disappear, as if I entered a relaxed state, and my heart rate and breathing became more stable.” Participants from the AF group also found auditory feedback effective but were willing to try adding visual feedback. One participant remarked: “The auditory feedback helped me relax, but adding visual feedback might help me be more certain about whether I’m in the right meditative state.”

6. Discussion

This study investigates the impact of kinesthetic interactive mindfulness training on emotion regulation, with a particular focus on the application of the “Digital Prayer Beads” device in stress relief. By integrating physiological indicators such as EDA and HRV, alongside STAI-S scores and qualitative data from participant interviews, this study highlights the potential of kinesthetic mindfulness devices in promoting emotion regulation and alleviating stress. This section will analyze the core research findings, conducting a comparative analysis with existing research in the fields of human-computer interaction, embodied mindfulness, and multimodal emotion regulation. It will highlight the differences and added value of this study in relation to previous literature, while also further advancing its theoretical contributions, design implications, and practical application value.

6.1. Effectiveness of kinesthetic interaction and its physiological mechanisms: From “distraction” to “embodied anchoring”

This study integrates kinesthetic interaction into mindfulness training, exploring the role of body movements in emotional regulation. The results demonstrate that hand-rotation-based kinesthetic interaction is an effective method for regulating emotions. Compared to natural recovery, participants in the AF and VAF groups who used the device showed significant reductions in STAI-S scores. Additionally, there was a notable decrease in the LF/HF ratio of HRV, along with a slower heart rate, confirming that guided and embodied mindfulness practices effectively promote a shift in the autonomic nervous system toward parasympathetic dominance, supporting Hypothesis H1.

The kinesthetic mindfulness device in this study creates a closed-loop feedback mechanism for emotional regulation through bodily interaction. Unlike traditional meditation techniques, kinesthetic interaction provides not only sensory feedback but also encourages participants to engage more actively and improve self-regulation during mindfulness practices. This advantage aligns with findings from Fartook et al. and Papadopoulou, who highlighted the importance of tactile and bodily perceptions in emotional regulation (Fartook et al., 2026; Papadopoulou, Paterson, 2025). However, both studies have limitations: they rely primarily on passive sensory feedback, lack active bodily movements in their designs, and seldom incorporate multidimensional physiological data to validate the intervention's physiological effects—gaps this study aims to fill.

Furthermore, the results advance our understanding of the underlying mechanisms, moving beyond the "attention distraction" theory. We introduce the "embodied anchoring" theory to explain its benefits: when stress causes attention to wander, regular hand movements coupled with synchronized feedback offer a stable, perceivable "anchor." As one participant described, "finger rolling helps maintain focus," illustrating this mechanism. This anchor helps shift cognitive resources from internal worries to immediate, controllable bodily sensations and external feedback, disrupting the cycle of anxiety. Physiologically, this results in reduced sympathetic nervous system arousal and enhanced parasympathetic activity. Indicators such as EDA and HRV show that kinesthetic feedback activates the parasympathetic system, encouraging a more relaxed physiological state. These findings align with previous research on mindfulness and the autonomic nervous system, complementing the work of Yu et al. (2021). In contrast to the passive tactile guidance of the ViBreathe system, this study emphasizes the crucial role of active, controllable hand kinesthetic involvement in regulating attention.

6.2. Tradeoffs in feedback modes: The balance between cognitive engagement and depth of regulation

The research findings suggest that audiovisual multimodal feedback is most effective in alleviating subjective anxiety. However, for certain physiological indicators of deep relaxation, such as RMSSD and SCL, VAF did not significantly outperform auditory feedback alone. This result reveals the inherent tradeoffs between different feedback modes, partially supporting hypotheses H2 and H3. Specifically, the VAF group demonstrated the lowest STAI-S scores, with users in interviews describing it as "intuitive" and "comprehensive," aligning with the expected effects of multimodal integration in enhancing information richness and immersion. This finding is consistent with the research of Liu and Gu, who found that audio-vibrational feedback enhanced the perception of virtual weight and increased interaction immersion, validating the positive role of multimodal feedback in improving user experience (H. Liu & Gu, 2026).

However, similar to the findings of Hao and Xie, this study reveals that multimodal feedback is not always the optimal choice in all contexts (Hou et al., 2025). Existing research predominantly emphasizes the positive value of multisensory integration, overlooking the potential negative impact of cognitive overload in high-pressure and high-anxiety situations. This study fills this gap with empirical data, uncovering the boundaries of multimodal feedback and its adaptability in different contexts.

Specifically, the AF group performed best in terms of RMSSD values reflecting parasympathetic nervous activity, and also had the lowest SCL values, indicating that minimal auditory feedback may be more suitable for deep physiological relaxation. Although the VAF group excelled in reducing subjective anxiety, some users noted that the visual content could cause mild distractions, thus diminishing the effectiveness of deep relaxation. This highlights the need for a balanced feedback mode in emotional regulation devices, tailored to the actual needs of users. This discovery challenges the traditional belief that multisensory integration is always the most optimal design approach. It also adds to the growing body of HCI research on the applicability of multimodal feedback in anxiety interventions, offering important insights for Park et al. 's haptic guidance system (Park et al., 2025). The study emphasizes the importance of balancing feedback richness with cognitive load in the design of emotional regulation devices, proposing the "goal-oriented feedback mode selection" principle. This principle serves as a crucial reference for the design of human-computer interactions in health-related applications, particularly for lightweight, low-load interaction optimizations in special emotional contexts such as anxiety and stress, further expanding the boundaries of multimodal emotional interaction research within the HCI field.

6.3. Exploratory observation of individual differences in experience and cultural preferences

In our interviews, we found that individual differences in experience and cultural preferences may have a potential impact on how users perceive and adapt to the digital prayer beads device.

The core experimental design and quantitative analysis of this study primarily focus on the stress-relief effects of different feedback modes. It did not incorporate standardized personality assessments, nor were cross-cultural control samples included, and no statistical models were used to test potential moderating effects. Therefore, the analysis in this section is based not on quantitative data validation, but on qualitative feedback from semi-structured interviews and exploratory observations, making it a preliminary set of insights.

Based on interview feedback, it is evident that different users exhibit varying preferences in their interactive experience and sensory responses. Some users are more receptive to multimodal audiovisual feedback, which provides rich information, while individuals who are sensitive to pressure are more affected by additional cognitive demands and tend to prefer simpler, quieter auditory cues. Furthermore, most participants prefer restrained, low-stimulation, and ritualistic hand-based interactions. This preference may stem from the fact that the participants were primarily young Chinese individuals, influenced by local mindfulness traditions and living environments. This finding aligns with the work of Chan et al. (2025), which explores "the impact of cultural factors on interaction experiences."

The above observations are specific to the current sample. This study is based on design inspiration drawn from a single cultural context (China) and does not involve cross-cultural validation, limiting the generalizability of the findings. Future research should refine the study design to further investigate how individual differences might influence the effectiveness of the intervention.

6.4. Haptic design and adaptive feedback

The preliminary study has demonstrated the potential of adaptive feedback in developing personalized interactions. The tactile materials used in the device received varied evaluations, suggesting that tactile feedback, as an emotional channel, is not universally effective. This observation aligns with the research of Huang et al. which found that tactile feedback can promote emotional regulation through emotional connections(X. Huang & Romano, 2025). However, the effectiveness of tactile feedback depends on its alignment with user preferences. This finding is also supported by the concept of "tactile adaptability" discussed by Karpodini et al. (2025). Building upon this foundation, this study introduces innovations and extensions. Currently, adaptive interaction and wearable physiological closed-loop control are crucial research areas within health-related HCI. For instance, Hirten et al. used PPG sensors to monitor HRV and implemented HRV-guided biofeedback interventions, which were instrumental in informing this study (Hirten et al., 2024). The motion-driven adaptive feedback system developed in this study differs from physiological signal-driven models by emphasizing a bidirectional dynamic adaptation between the user's actions and the system's feedback. This approach better aligns with the core principle of active participation in mindfulness training. Moreover, it incorporates a multimodal feedback adaptive switching mechanism, expanding the regulation capabilities of traditional tactile feedback.

The adaptive dynamic tactile mechanism developed in this study has begun to show its value by creating a basic personalized interaction. This adaptive tactile feedback has been shown to activate the parasympathetic nervous system, effectively lowering heart rate and skin conductance responses, helping individuals achieve a relaxed state. Compared to traditional unidirectional feedback, adaptive feedback allows interactions to be fine-tuned based on the user's real-time motions, enabling bidirectional interaction. This enhances both the efficiency of emotional regulation and the naturalness of the experience.

6.5. Research practice, policy recommendations, and industry value

This study's core findings contribute significant theoretical innovation and offer practical, transferable design principles for future applications. These insights promote the translation of kinesthetic mindfulness interactions into health-related HCI fields. From a design practice perspective, the study explores how to integrate Eastern mindfulness traditions with modern multimodal interaction, providing valuable references for both the digitalization of intangible cultural heritage and the development of mental health products. This integration not only addresses the call for culturally diverse designs but also enriches the conceptualization of physiologically driven adaptive systems, further advancing the nuanced development of health-related HCI products.

The following transferable design principles for emotional regulation devices have emerged from this study:

1. Kinesthetic Interaction and Emotional Regulation: Designs should focus on enhancing users' engagement in emotional regulation through body movement feedback, particularly in high-pressure environments, offering low-load, non-intrusive tools for emotional regulation.
2. Subtle Use in Public Spaces: Devices should be designed to be discreet in public environments, ensuring that their use does not disrupt work or social interactions while still effectively promoting emotional regulation.
3. Balancing Multimodal Feedback and Cognitive Load: When designing multimodal feedback systems, the cognitive load on users must be considered. Adaptive options should be available to prevent excessive information from overwhelming the user, ensuring feedback remains effective and engaging.

These principles provide a theoretical foundation for the design of future emotional regulation devices, particularly in the context of cross-cultural environments and diverse application settings.

In the field of public health, the device explored in this study—being a non-pharmacological, low-load intervention tool—has significant potential for widespread use, especially in campus mental health centers and corporate wellness programs. In comparison with existing meditation apps, the digital kinesthetic mindfulness tool proposed here offers differentiated interaction modes, which enhance the accessibility of mental health services and address the psychological health goals outlined in the "Healthy China" strategy.

At the policy level, we recommend the following initiatives:

1. Mental Health Education Reform in Universities: Integrate kinesthetic mindfulness devices into mental health curricula to help students develop effective emotional regulation strategies.
2. Optimizing Corporate Health Management Policies: Encourage high-stress industries to provide employees with similar tools for effective emotional regulation.
3. Supporting the Cultural and Technological Integration Sector: Establish dedicated funding to support innovative projects that fuze traditional cultural elements with modern technology, promoting advancements in relevant fields.
4. Advancing Cross-Cultural Research Funding: Fund cross-cultural studies to deepen global health HCI data, aiding health interventions worldwide.

In both industry and research, this study responds to the growing trends of lightweight, embodied, and culturally localized health HCI designs, particularly by contributing to data on emotional regulation technologies within Eastern cultural contexts. Through interdisciplinary collaboration, we have facilitated the convergence of design, psychology, medicine, and computer science. The "Embodied Anchoring" mechanism proposed in this study introduces a fresh perspective to embodied interaction research and offers valuable insights for both academic development and practical applications in related fields.

6.6. Limitations

While this study provides empirical support for the role of kinesthetic mindfulness and adaptive feedback in emotion regulation, several limitations must be acknowledged. First, the sample is relatively homogenous, as all participants are from China, and this lack of cultural diversity may limit the generalizability of the findings. Future studies should expand the sample size and include cross-cultural participants to enhance external validity. Second, the intervention and training period in this study were brief, focusing primarily on the immediate effects of the system. This limits the ability to assess the system's stability and long-term impact, which could be addressed through longitudinal studies in future research. Third, the experiment was conducted in a controlled laboratory setting, which, while enhancing internal validity, may reduce ecological validity. Future research could assess the system's performance in real-world work or study environments. Additionally, this study primarily examines the effects of kinesthetic adaptive feedback and does not compare the synergistic effects of multimodal feedback. Future studies could introduce visual, auditory, and tactile feedback under controlled cognitive load. Lastly, there remains a need to explore how to customize feedback more precisely based on individual differences, such as personality traits, emotional states, and cultural backgrounds.

7. Conclusion

This study investigates the application of kinesthetic interaction technology in emotion regulation, introducing the embodied anchoring effect triggered by hand-based kinesthetic interactions to aid users in alleviating stress and anxiety. The findings suggest that kinesthetic interaction integrates bodily movements into the emotion regulation process, providing a more intuitive and immersive approach that fosters the restoration of psycho-physiological balance. Unlike traditional static meditation techniques, kinesthetic interaction emphasizes bodily engagement, transitioning emotion regulation technology from a narrow focus on psychological or physiological interventions to a more interactive and personalized experience. The central contribution of this research lies in offering a novel perspective on emotion regulation technology through kinesthetic interaction and embodied anchoring effects. This is particularly relevant for addressing the growing demand for personalized, customizable designs and cross-cultural adaptation, providing a solid foundation for future studies and applications in this field.

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Author contributions

CRediT: **Haoyu Zhang**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **Xiaoying Li**: Funding acquisition, Resources, Supervision.

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Data availability statement

The datasets generated or analyzed during this study are available from the corresponding author on reasonable request.

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